
Building Reliable Long-Form Generation via Hallucination Rejection Sampling

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Abstract

Large language models (LLMs) have achieved remarkable progress in open-ended text generation, yet they remain prone to hallucinating incorrect or unsupported content, which undermines their reliability. This issue is exacerbated in long-form generation due to hallucination snowballing, a phenomenon where early errors propagate and compound into subsequent outputs. To address this challenge, we propose a novel inference-time hallucination mitigation framework, named Segment-wise HALLucination Rejection Sampling (SHARS), which uses an arbitrary hallucination detector to identify and reject hallucinated segments during generation and resample until faithful content is produced. By retaining only confident information and building subsequent generations upon it, the framework mitigates hallucination accumulation and enhances factual consistency. To instantiate this framework, we adopt semantic uncertainty as the detector and introduce several vital modifications to address its limitations and better adapt it to long-form text. Our method enables models to self-correct hallucinations without requiring external resources such as web search or knowledge bases, while remaining compatible with them for future extensions. Empirical evaluations on standardized hallucination benchmarks demonstrate that our method substantially reduces hallucinations in long-form generation while preserving or even improving the informativeness of generation. Code is available at: <https://github.com/TreeLLi/hallucination-rejection-sampling>.

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1. Introduction

Large language models (LLMs) (OpenAI, 2025; Yang et al., 2025a; Grattafiori et al., 2024) have markedly expanded the frontiers of artificial intelligence, demonstrating impressive capabilities in open-ended text generation across domains such as question answering (Min et al., 2023; Wei et al., 2024), code synthesis (Jimenez et al.), and scientific research (Lu et al., 2026). However, their practical deployment is hindered by a persistent and well-documented challenge: hallucination (Ji et al., 2023). Hallucinations arise when models generate content that is factually inaccurate, unsupported, or in conflict with the provided input (Bang et al., 2025), often delivered with high fluency. This phenomenon undermines the reliability of model output and user trust, and poses risks in high-stakes applications like healthcare and medicine.

Hallucinations are particularly concerning in open-ended generation, where the extended and unconstrained nature of the outputs makes it especially challenging to validate. In addition, prior studies (Zhang et al., 2024; Zhao et al., 2025; Yang et al., 2025b) have shown that longer generations tend to amplify hallucination risk, a phenomenon known as *hallucination snowballing*, in which early errors propagate and trigger additional mistakes. This underscores the importance of intervening early in the generation process to interrupt error accumulation and thereby reduce hallucinations.

Separately, a growing body of research (Wei et al., 2022; Yao et al., 2023; Muennighoff et al., 2025; DeepSeek-AI et al., 2025) has investigated the paradigm of inference-time compute scaling, which improves model performance by allocating additional computation at generation time. This paradigm is particularly well-suited for hallucination mitigation in high-stakes domains such as healthcare, scientific discovery, and law, where users are often willing to accept slower responses in exchange for more factual and reliable outputs. Nevertheless, this direction remains underexplored, and to the best of our knowledge, there are no well-established findings on how inference-time scaling affects factuality in open-ended generation.

Inspired by these insights, we introduce a general inference-time compute framework, termed Segment-wise HALLucination Rejection Sampling (SHARS), to mitigate hallucinations in open-ended generation. SHARS

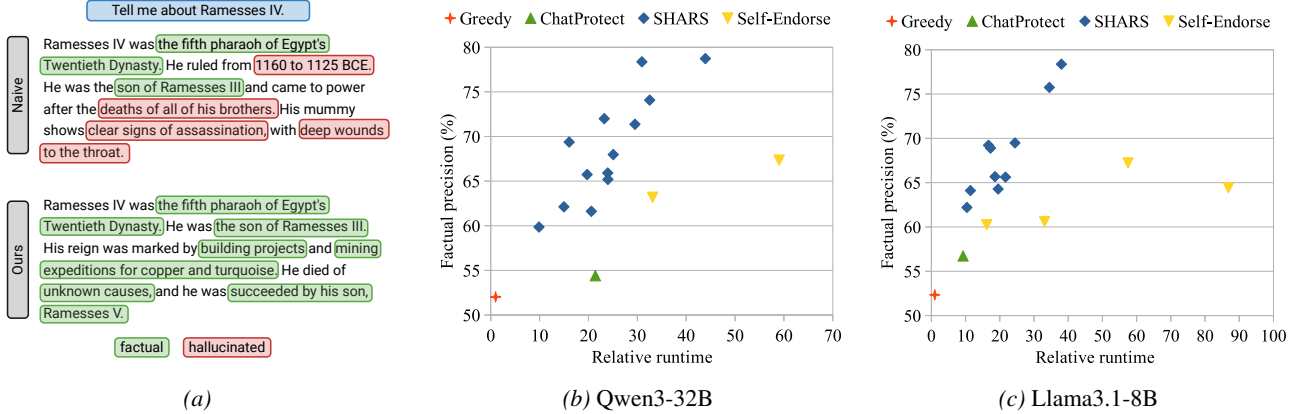


Figure 1. (a) Comparison of biographies generated by Greedy decoding and our method. Unlike Greedy decoding, our method rejects hallucinated content, preserves factual information, and acquires additional factual content (the last two sentences in the displayed generation) beyond the original information space. (b) and (c) Scaling of factual precision with respect to inference-time computation on the FactScore benchmark (Min et al., 2023). Inference-time computation is approximated by relative runtime, measured as a factor of the runtime of the corresponding Greedy decoding method for each setup. Full experimental details are provided in Section 5.

leverages an arbitrary detector to identify and reject hallucinated content as it is produced during generation, preserves only factual segments, and builds subsequent outputs upon them (Figure 1a). This design aims to increase the proportion of factual information in the final output while disrupting hallucination snowballing from its early stages.

To instantiate this framework, **we further propose a new hallucination detection method, HalluSE, tailored for long-form generation.** HalluSE builds upon the prior semantic entropy approach (Farquhar et al., 2024), incorporating several refinements to address its limitations and improve detection effectiveness. **Notably, SHARS is designed to be detector-agnostic, allowing it to integrate with any hallucination detection method** and thereby broadly benefit from future advances in hallucination detection research.

We conduct extensive experiments on diverse long-form factuality benchmarks, including FactualBio (Farquhar et al., 2024), FactScore (Min et al., 2023), and LongFact (Wei et al., 2024), to evaluate our methods. Empirical results show that HalluSE significantly improves hallucination detection accuracy over prior approaches in long-form generation. SHARS further proves effective in mitigating hallucinations in open-ended generation while preserving, and in some cases enhancing, output informativeness. Importantly, **SHARS exhibits a promising scaling property: when appropriately configured, factuality continues to improve as additional inference-time computation is allocated within a certain range** (Figures 1b and 1c). For instance, SHARS improves factual precision by about 26% for evaluated models on the FactScore benchmark.

2. Related Works

Hallucination mitigation. To mitigate hallucinations, Tian et al. (2024) fine-tune models using preference data generated from a retrieval-enabled judge and direct preference optimization (DPO), enabling the model to prefer factual responses. Huang & Chen (2024) propose FactAlign, which assigns sentence-level factuality rewards to reinforce supported spans in long-form outputs. Gu et al. (2025) introduce Mask-DPO, which masks non-factual sentences during preference optimization so that updates focus exclusively on factual content.

Inference-time mitigation approaches have also been explored. Integrative decoding (Cheng et al., 2025b) aggregates self-consistent continuations by jointly selecting supported tokens. Chuang et al. (2024) propose DoLa, which reweights next-token probabilities by contrasting logits from late and early layers. Retrieval-augmented generation (RAG) (Lewis et al., 2020) grounds generation on retrieved passages to replace unsupported spans, while Cai et al. (2024) improve RAG by introducing outline-guided generation and factuality-aware optimization for web-augmented long-form outputs. More recently, Cheng et al. (2025a) incorporate tree search-based algorithms to enable explicit slow-thinking generation, mitigating hallucinations during inference.

Hallucination detection. Farquhar et al. (2024) introduced hallucination detection via semantic entropy, which estimates uncertainty in the space of meanings by clustering diverse model samples and measuring entropy over the induced semantics. They benchmarked this method against two alternatives: Self-Check (Manakul et al., 2023), where the model verifies its own assertions, and P(True), which measures the probability that the model predicts the token

“True” when few-shot prompted to compare a main answer with alternatives. Another training- and retrieval-free approach by Mündler et al. (2024) detects hallucinations by eliciting multiple responses and identifying contradictions or inconsistencies. Other methods train lightweight probes. For instance, Kossen et al. (2024) trained probes to approximate semantic entropy from hidden states of a single generation, while Obeso et al. (2025) trained probes on web-search-grounded, entity-level labels to detect hallucinations in real time. Alternatively, Min et al. (2023), Wei et al. (2024), and Zhao et al. (2025) detect hallucinations by decomposing generated text into atomic facts and checking them against trusted external sources.

3. SHARS: Segment-wise HALLUCINATION Rejection Sampling

Motivation. We observe that open-ended questions often admit an effectively infinite range of relevant information that can constitute a valid answer, yet in practice models draw on only a limited subset of this space when generating responses. Intuitively, if hallucinated content in the initial generation can be filtered out and the model is guided to explore the remaining information space for truthful content to fill these gaps, the resulting generation can be free of hallucinations. Moreover, by dynamically grounding generation on truthful information, this process could potentially disrupt the error compounding caused by earlier mistakes and increase the likelihood of sampling factual content.

3.1. Segment-wise Hallucination Rejection Sampling

Following this motivation, we propose our general inference-time compute framework, SHARS, which leverages an arbitrary detector to identify and reject hallucinated content during generation. SHARS partitions the generation into multiple segments and applies hallucination rejection sampling sequentially as each sentence is produced. For a given sentence, hallucination rejection sampling invokes a hallucination detector to assess its factuality. Based on the detection outcome, the sentence is either (i) discarded if it contains no factual information, (ii) rewritten to remove hallucinated content if it mixes factual and hallucinated information, or (iii) retained if it is entirely factual, with no hallucinations detected. Generation terminates when one of the following occurs: (1) an end-of-sequence (EOS) token is sampled; (2) the maximum new-token budget is reached; or (3) fully hallucinated sentences are sampled in N consecutive attempts. The full procedure is summarized in Algorithm 1.

Our method differs from conventional rejection sampling, also known as the best-of- N sampling, for inference-time scaling in three key aspects. First, rejection sampling is performed in a segment-wise and dynamic manner rather

Algorithm 1 Pseudocode of SHARS.

Data: User query q

Result: Verified response to the user query

verified_text, hallued_text $\leftarrow$ “”, “”

while not End_Of_Sequence do

 sent $\leftarrow$ next_sent(M , q , verified_text, hallued_text)

 verified_facts, hallued_facts $\leftarrow$ detect_hallu(M , sent, verified_text)

if len(verified_facts) = 0 **then**

 hallued_text $\leftarrow$ hallued_text + sent

else

if len(hallued_facts) $\neq$ 0 **then**

 sent $\leftarrow$ rewrite_sent(M , q , verified_facts)

 verified_text $\leftarrow$ verified_text + sent

 hallued_text $\leftarrow$ “”

 // break if no verified_facts for N times in a row

return verified_text

than applied once to the entire generation. Second, we sample one candidate sentence at a time and resample only when the current sentence is rejected, instead of generating multiple candidates simultaneously. Third, in cases where a sentence contains both factual and hallucinated information, we rewrite it to remove hallucinations rather than discarding it entirely. The latter two strategies improve efficiency and make it more practical for inference-time deployment.

Hallucination detector. SHARS is designed to operate with any detector by treating the hallucination detector as a black box. In this work, we instantiate SHARS with our entropy-based HalluSE detector (technical details are described later in Section 4.3). We adopt HalluSE because (1) it does not require training a new probe model to detect hallucinations like Obeso et al. (2025), and because (2) it does not rely on external tools or reference knowledge sources. These properties allow seamless integration into SHARS and enable zero-shot application across new domains.

We acknowledge that hallucination detection is an active research area, and many alternative methods (Aichberger et al., 2026; Duan et al., 2024; Manakul et al., 2023; Chen et al., 2024) may be suitable for integration into our framework. We leave this exploration for future work, as our current choice already achieves substantial reductions in hallucinations and significant improvements in factual precision compared with existing state-of-the-art mitigation methods, as shown in Section 5.1.

HalluSE estimates the uncertainty of the knowledge probed by generated questions and uses it as a proxy for the uncertainty of the corresponding fact. For example, consider the fact to be verified: Alan Turing is an athlete. The relevant knowledge in this case is Alan Turing’s profession. If the model is uncertain about this knowledge, it sug-

Algorithm 2 Pseudocode of next sentence sampling.

Data: q , verified_text, hallued_text

Result: A new sentence

text_sofar $\leftarrow$ "" input $\leftarrow q$ + verified_text + hallued_text

while True **do**

 token $\leftarrow$ next.token(M , input)

 text $\leftarrow$ decode(token)

 text_sofar $\leftarrow$ text_sofar + text

 sents $\leftarrow$ split_sents(text_sofar)

if len(sents) ≥ 2 **then**

 sent $\leftarrow$ sents[0]

break

 input $\leftarrow$ input + text

return sent

gests that not only the underlying fact is likely hallucinated, but also that alternative sampled facts for the profession are likely hallucinated. While this improves the efficiency of hallucination detection, it also raises a challenge for sentence sampling: how do we generate a new sentence with knowledge distinct from that in the hallucination?

Sentence sampling. To address the above challenge, we explore two strategies, termed Temperatures and Following. The Temperatures strategy gradually increases the decoding temperature for sampling a new sentence as the number of consecutive hallucinated sentences grows. In other words, the longer the model is stuck at a given point in generation, the more randomness is introduced to encourage exploration of alternative continuations. This approach leverages the model’s inherent stochasticity to produce diverse sentences, but it can be less efficient as it does not explicitly incorporate information from the identified hallucinated sentences.

In contrast, the Following strategy temporarily retains the identified hallucinated sentences in the generation and samples the next sentence by continuing the generation process, as illustrated in Algorithm 2. This leverages the model’s inherent content planning ability to reduce the likelihood of repeatedly generating content about the same knowledge. For example, a model will typically avoid generating a second birthday for an individual once one has already been stated. However, this approach risks allowing hallucinations to influence subsequent generation. To mitigate this effect, we clear the pool of hallucinated sentences whenever new factual information is identified and retained, preventing the pool from becoming excessively large, as shown in Algorithm 1. Furthermore, hallucinated sentences are used solely for sentence sampling and are not passed to HalluSE as context for computing semantic entropy, ensuring that existing hallucinations do not affect the identification of hallucinations in newly sampled sentences. The Following strategy is ultimately adopted due to its superior empirical performance, as discussed in Section 5.7.

Sentence rewriting. We employ an LLM to rewrite the sentence to remove its hallucinated content while preserving factual information. Specifically, we provide the LLM with a list of factual claims identified by HalluSE and prompt it to generate a sentence comprising those claims, rather than supplying the original sentence along with hallucinated claims and asking it to remove them. Empirically, we find that the former approach performs better with small- to medium-scale models such as Qwen3-32B, Llama3.1-8B, and even Qwen3-4B-Instruct. We hypothesize that this advantage arises because LLMs are more effective when guided by positive examples than by negative ones.

The rewriting LLM can be any model with sufficient instruction-following capability to perform the task. In this work, we use the same model as the main generation model. The rewriting prompts are provided in Appendix A.

3.2. Abstention Mechanism

Our third termination condition leads to a novel dynamic abstention mechanism based on the model’s parametric knowledge and internal confidence. Assuming sufficient diversity in sentence sampling, our method abstains after generating N fully hallucinated sentences covering different aspects of the user query in a row. This abstention may occur either at the outset or midway through a generation, with the latter case allowing the model to first produce information it is confident is factual.

4. HalluSE: Detecting Hallucinations in Long-Form Generation

This section introduces HalluSE, our uncertainty-based hallucination detection method for long-form text generation. HalluSE builds on the prior semantic entropy approach for long-form generation (Farquhar et al., 2024), while addressing several of its key limitations.

4.1. Background: Semantic Entropy

Semantic entropy (Farquhar et al., 2024) is an uncertainty measure that captures the variability of a model’s predictions in the semantic space rather than the token space. Instead of only considering surface-level probability distributions over tokens, semantic entropy groups candidate generations into meaning-equivalent clusters and measures the entropy across these clusters. Given a set of candidate generations sampled from the model, each generation is mapped into a semantic cluster C_i . The probability of a cluster is defined as the sum of token-level probabilities of all generations assigned to it: $p(C_i) = \sum_{y \in C_i} p(y)$, where $p(y)$ is the model probability of generation y . The semantic entropy is then computed as the entropy over cluster probabilities: $H_s = -\sum_i p(C_i) \log p(C_i)$. The full technical details can

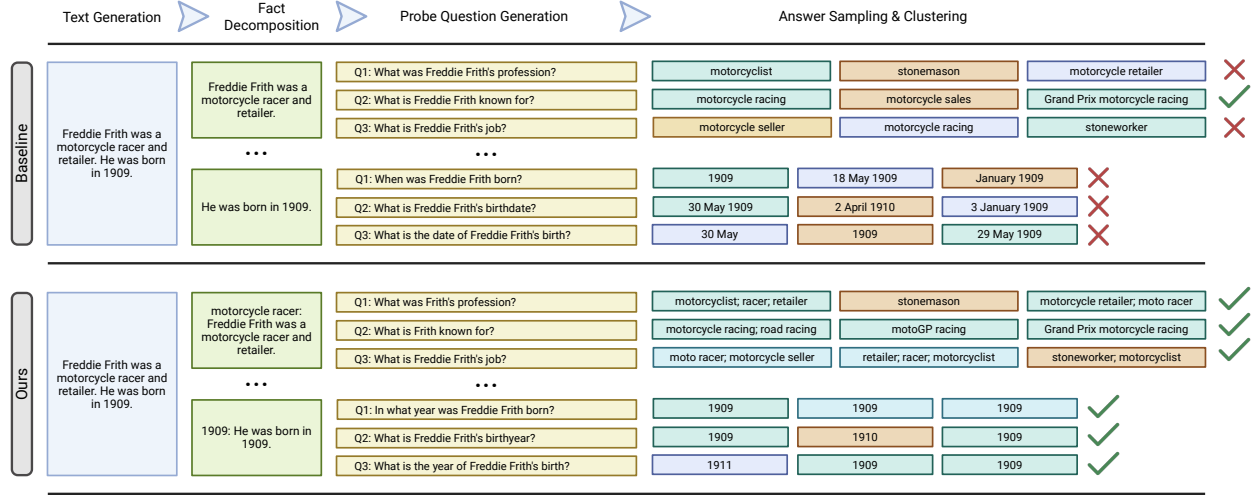


Figure 2. Illustration of naive long-form semantic entropy method and our proposed HalluSE. Different colors under ‘Answer Sampling & Response’ denote distinct semantic clusters of generated responses. A green check indicates low semantic entropy (high agreement, reliable answers), while a red cross marks high semantic uncertainty (likely hallucinated content).

be found in Farquhar et al. (2024).

Low semantic entropy indicates semantic agreement among candidate generations, while high semantic entropy reflects semantic disagreement and is often associated with hallucinations. This property makes semantic entropy a natural signal for hallucination detection: when the model is confident and semantically consistent, the likelihood of hallucination is lower, whereas high semantic entropy often correlates with unsupported or erroneous content.

4.2. Naive Long-Form Semantic Entropy

The semantic entropy method described above assumes that candidate answers are short-form. To extend it, Farquhar et al. (2024) proposed a naive approach for applying semantic entropy to long-form generation. As shown in Figure 2, the generation is first decomposed into a set of fact claims. For each fact claim, several probe questions with expected answers are generated to query the fact, and the short-form semantic entropy method is then applied to each question. This procedure effectively reduces long-form hallucination detection to a series of short-form detection tasks.

The naive long-form semantic entropy method faces two main limitations as illustrated in Figure 2. First, it decomposes a generation into fact claims without distinguishing which entity within each claim should be validated. This ambiguity can cause the wrong entity to be probed downstream. For example, given the query Tell me about Alan Turing and the generation Alan Turing is a computer scientist, the entity of interest is clearly computer scientist rather than Alan Turing. However, the prior method may in-

correctly generate probe questions such as Who is a computer scientist?.

Second, the naive approach assumes that each probe question has only a single valid answer, so any uncertainty in sampled answers is attributed solely to the model. In practice, however, probe questions can admit multiple valid answers. For example, in biographies, a prominent individual may hold multiple professions. A probe question such as What is XX’s profession? may thus have several correct answers. Even if the model consistently samples correct but different professions, the resulting semantic entropy remains high, incorrectly flagging the fact as hallucinated.

4.3. HalluSE

HalluSE addresses the limitations of the naive long-form semantic entropy method through three key refinements as illustrated in Figure 2. First, it decomposes each generation into pairs of entities and fact claims. Second, it improves the prompting strategy with clearer instructions, structured formatting, and few-shot examples. In particular, HalluSE guides the LLM to generate probe questions with unambiguous expected answers, thereby reducing unnecessary cases of multiple valid answers. Third, it explicitly instructs the LLM to provide all valid answers, when applicable, in each sampling step. The complete HalluSE pipeline is as follows:

1. **Fact Decomposition:** given a generation, HalluSE decomposes it into a set of facts, where each fact consists of an entity and a claim describing a piece of atomic information about that entity from the model response.
2. **Question Generation:** for each fact, HalluSE gener-

Algorithm 3 Pseudocode of HalluSE.**Data:** text, c , M , Q , A , θ **Result:** verified facts, hallucinated factsverified_facts, hallued_facts $\leftarrow$ [], []facts $\leftarrow$ decompose_facts(M , text)**for** ($entity$, $claim$) **in** facts **do** questions $\leftarrow$ gen_questions(M , Q , $entity$, $claim$) $H_s \leftarrow$ [] **for** $question$ **in** questions **do** $H_s \leftarrow H_s \cup$ semantic_entropy(M , A , c , question) $H_s \leftarrow$ mean(H_s) **if** $H_s < \theta$ **then** verified_facts $\leftarrow$ verified_facts $\cup$ ($entity$, $claim$) **else** hallued_facts $\leftarrow$ hallued_facts $\cup$ ($entity$, $claim$)**return** verified_facts, hallued_facts

ates Q probe questions in which the entity and claim serves as the expected short-form and long-form answer, respectively.

3. **Answer Sampling:** for each probe question, it produces A answers conditioned on the preceding context c , i.e., the text appearing before the fact in the response.
4. **Semantic Entropy Computation:** semantic entropy is computed from the sampled answers per question and averaged across the Q questions, yielding the semantic entropy of the fact.
5. **Hallucination Identification:** A fact is classified as hallucinated if its semantic entropy exceeds a predefined threshold θ ; otherwise, it is deemed factual.

The full procedure is summarized in Algorithm 3. Fact Decomposition, Question Generation, and Answer Sampling are implemented by prompting a pretrained instruction-following LLM, denoted as M , with the specific prompts detailed in Appendix A. Semantic Entropy Computation is implemented with its discrete formulation (Farquhar et al., 2024). The LLMs for Fact Decomposition and Question Generation can be arbitrary, while the LLM for Answer Sampling should match the model used to produce the given response. In this work, we employ the same model for all components, including response generation.

5. Results

Experiment setup. We mainly evaluate our method on the FactScore benchmarks using Qwen3-4B, Qwen3-32B (Yang et al., 2025a) and Llama3.1-8B-Instruct (Grattafiori et al., 2024). Qwen3-4B/32B follows officially recommended decoding settings with temperature 0.7, top- k 20, and top- p 0.8, while Llama3.1-8B-Instruct uses temperature 0.7, top- k

Table 1. Performance of baselines and our methods on the FactScore benchmark without constraints on response length. The best score for each metric is highlighted in **bold**.

Method	Resp. (%)	Unsup.	Supp.	FPrec. (%)
Qwen3-4B				
Greedy	91.2	11.2	11.2	50.0
DoLa	94.5	11.1	11.9	51.8
ChatProtect	91.8	9.7	10.8	52.6
Self-Endorse	92.3	6.6	9.2	58.2
Ours-Resp	92.9	8.8	14.9	63.0
Ours-Info	89.0	8.2	16.3	66.6
Ours-Prec	69.8	5.7	16.2	74.0
Llama3.1-8B				
Greedy	99.5	5.7	6.7	53.7
DoLa	99.5	5.7	6.7	53.8
ID	98.3	5.0	5.7	53.5
ChatProtect	97.3	5.0	6.5	56.7
Self-Endorse	96.7	4.1	6.3	60.6
Ours-Resp	99.5	3.2	5.7	64.1
Ours-Info	88.5	1.9	5.9	75.6
Ours-Prec	78.6	1.4	5.0	78.4
Qwen3-32B				
Greedy	99.5	8.8	9.7	52.4
DoLa	95.6	9.3	8.2	53.1
ChatProtect	98.9	8.1	6.8	54.4
Self-Endorse	91.8	4.9	8.4	63.2
Ours-NE	97.3	4.2	9.9	70.1
Ours-Resp	97.8	5.7	11	65.7
Ours-Info	92.9	4.2	11.7	73.5
Ours-Prec	82.4	3.1	11.1	78.4

50, and top- p 0.9. For baselines, we use Greedy decoding, DoLa (Chuang et al., 2024), ID (Cheng et al., 2025b), ChatProtect (Mündler et al., 2024), and Self-Endorse (Wang et al., 2024). The ID results for Qwen3 models are omitted because the authors’ released code is incompatible with Qwen3. The full experiment setup and the configuration of our methods are given in Appendix B.1.

Factual precision (“FPrec.”) is defined as the proportion of supported claims (“Supp.”) relative to the total number of claims (“Supp.” + “Unsupp.”). Response rate (“Resp.”) denotes the proportion of queries answered without refusal. Factual precision and the number of fact claims are computed with generations that answer without refusal.

For each model, results of our method are reported under three hyperparameter settings: Ours-Resp maximizing the response rate, Ours-Info maximizing the number of supported claims, and Ours-Prec maximizing factual precision. In addition, we also replace semantic entropy (SE)

Table 2. Performance of our methods for Qwen3-32B on the FactScore benchmark with a 200-word response length constraint. Models are prompted to generate around 200 words, which exceeds the average length produced without such constraint.

Method	Resp. (%)	Unsup.	Supp.	FPrec. (%)
Greedy	98.9	16.2	22.4	58.0
DoLa	97.8	16.9	22.3	56.9
ChatProtect	97.8	14.7	21.3	59.2
Ours-Info	98.4	11.8	29.1	71.0
Ours-Prec	84.6	6.7	23.6	77.9

with token-level naive entropy (NE) to measure uncertainty, named Ours-NE, while keeping the rest of the pipeline unchanged.

5.1. Reduced Hallucinations & Increased Information

Reduced hallucination rate. As shown in Tables 1 and 2, our method substantially reduces hallucination rates across different models and generation lengths. It consistently improves factual precision over the Greedy baseline by approximately 20–26% and significantly decreases the number of unsupported fact claims that are hallucinated by the model.

Although NE underperforms SE (70.1% vs. 78.4% FPrec.), it still surpasses the strongest baseline, Self-Endorse (63.2% FPrec), indicating that SHARS’ effectiveness stems from both the overall framework and the improved detector.

Increased factual information. In addition, Tables 1 and 2 show that our method increases the number of supported fact claims across all setups with Qwen3-32B, indicating that the generated responses contain more factual information and are thus more helpful. For Llama3.1-8B-Instruct, our method slightly reduces supported fact claims, but this is minor compared to the substantial reduction in hallucination.

Abstention. We observe that our method achieves the highest factual precision and the largest number of supported facts, albeit with a lower response rate. This indicates that the method effectively identifies user queries for which the underlying model has limited knowledge and abstains from answering. To further validate its effectiveness in mitigating hallucinations independent of additional abstention, we report results under a matched response rate with the baseline, denoted Ours-Resp in Table 1 and Ours-Info in Table 2. Even under this setting, our method substantially improves factual precision compared to the baseline.

We further examine factual precision (risk) as a function of response rate (coverage) by varying the uncertainty threshold for detection in Figure 3. We observe a clear and smooth trade-off. This demonstrates that the gains are not solely due to extreme abstention, but reflect a controllable trade-off

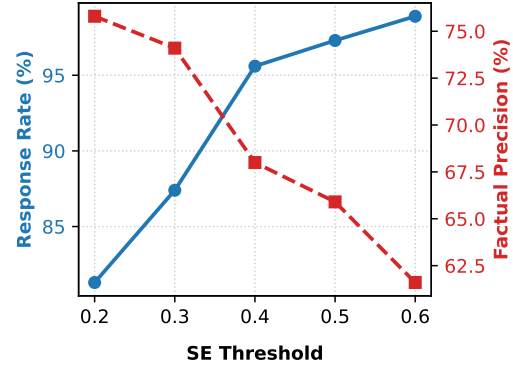


Figure 3. The change of response rate and factual precision with the semantic entropy threshold θ .

Table 3. Performance of combining FactAlign with our method on the FactScore benchmark for the Llama3-8B-Instruct model.

Method	Resp. (%)	Unsup.	Supp.	FPrec. (%)
No Response Length Constraint				
FactAlign	100.0	4.2	4.7	53.1
+ Ours-Resp	98.4	1.7	3.7	69.1
+ Ours-Prec	73.6	0.9	3.6	80.6
200-word Response Length Constraint				
FactAlign	100.0	13.6	16.9	55.4
+ Ours-Resp	100.0	8.6	14.6	62.9
+ Ours-Prec	78.6	4.0	15.3	79.1

where reliability improves consistently as lower-confidence outputs are filtered. Importantly, even in the high-coverage regime (Resp. > 95%), SHARS still achieves substantial improvements, indicating that gains persist beyond selective non-response. Moreover, since abstention is triggered by repeated failures to generate verified content, this behavior suggests that SHARS naturally abstains on queries where the model lacks reliable knowledge, rather than arbitrarily rejecting responses.

To assess consistency, we repeat the experiments for Qwen3-32B and Llama3.1-8B in Table 1 three times and measure the mean and standard deviation in Table 7 in Appendix C.1, which confirm the robustness and consistency of our results.

5.2. Generalization to Small-Scale Models

To further validate effectiveness on smaller models, we additionally evaluated our method using Qwen3-4B. It achieved comparable gains in factual precision (+16–24%), as shown in Table 1. We note that models with more than 4B parameters already cover a wide range of architectures used in both academic research and real-world applications. These results indicate that, although SHARS involves multiple LLM calls, it does not rely heavily on strong instruction-following capabilities. In other words, the effectiveness of the method

Table 4. Performance of the baseline and our methods on the FactualBio benchmark with Qwen3-32B when detecting both Major and Minor hallucinations.

Method	AUROC	AURAC	Acc@0.8	Acc@0.9
Self-Check	57.6	69.3	73.5	73.5
P(True)	69.8	73.3	70.0	70.0
Naive SE	66.2	73.1	70.5	70.5
Ours	72.9	77.3	75.4	72.8

does not diminish for smaller or less capable models.

5.3. Complementary to Training-Time Methods

The results in Table 3 show that our method provides strong complementary benefits to the training-time hallucination mitigation method FactAlign (Huang & Chen, 2024). Under both unconstrained and length-constrained settings, adding Ours-Resp consistently reduces unsupported claims and improves factual precision, while Ours-Prec achieves the largest gains—boosting precision from 53.1% to 80.6% without length constraints and from 55.4% to 79.1% with a 200-word limit. These results indicate that combining FactAlign with SHARS produces a more effective and reliable hallucination-mitigation strategy than FactAlign alone.

5.4. Efficient Scaling of Factual Precision

Across both figures in Figures 1b and 1c, **SHARS consistently demonstrates a far better accuracy–efficiency trade-off than the baselines**. In the Qwen3-32B plot, SHARS points cluster in the region of relatively low runtime (roughly 10–40x) while achieving higher factual precision (around 60–78%). In contrast, Self-Endorse requires much higher runtime (about 35–60x) to reach similar or even lower precision, and ChatProtect incurs additional computational cost with only limited improvement over the Greedy baseline. The trend is even more pronounced for Llama3.1-8B: SHARS maintains strong precision (around 68–78%) at runtimes below 50x, whereas Self-Endorse pushes beyond 60x runtime to achieve comparable accuracy. Taken together, the results show that SHARS delivers higher precision at substantially lower computational cost, making it significantly more efficient than existing methods.

5.5. Long-Form Hallucination Detection

This section compares HalluSE against long-form semantic entropy and other closely related detection baselines.

Experiment setup. We evaluate our hallucination detection method **without SHARS applied** for Qwen3-32B (Yang et al., 2025a) on the FactualBio dataset introduced by Farquhar et al. (2024). FactualBio contains paragraph-length biographies of 21 individuals sampled from the WikiBio

Table 5. Performance of the baseline and our methods on the LongFact benchmark under different generation length constraints.

Method	Resp. (%)	Unsup.	Supp.	FPrec. (%)
No Response Length Constraint				
Greedy	100.0	1.7	23.1	93.0
Ours	100.0	1.1	21.2	94.6
200-word Response Length Constraint				
Greedy	100.0	3.2	43.4	93.0
Ours	100.0	2.5	41.8	94.4

dataset (Lebret et al., 2016). Each paragraph-length biography in the FactualBio dataset is broken down into individual sentences, which are labeled True, Incorrect-Minor, or Incorrect-Major, depending on the severity of the false claim. For example, the claim that an individual was knighted, though they were not, is considered Incorrect-Major, while a reported birthdate in the wrong month is considered Incorrect-Minor. The same answers generated by GPT-4 were evaluated for different detection methods.

To benchmark our hallucination detection method, we extended the FactualBio dataset to include “entities”, with respect to which our method evaluates the semantic uncertainty of each claim. The Self-Check baseline, rather than evaluating semantic uncertainty, simply asks the LLM whether the factoid is likely to be true. The P(True) baseline considers the probability that the LLM predicts that the next token is “True” when few-shot prompted to compare the original answer with plausible alternatives.

AURAC summarizes how much accuracy improves when discarding the most uncertain answers. Accuracy@0.8/0.9 report the model’s accuracy after discarding the top 20% and top 10% most uncertain responses, respectively.

Improved detection accuracy. In Table 4, we observe that our method improves hallucination detection AUROC significantly. AUROC measures how well the method distinguishes correct from incorrect answers across all thresholds.

5.6. Additional Results on LongFact benchmark

In addition to FactScore, we evaluate our method on an alternative long-form factuality benchmark, LongFact (Wei et al., 2024). As shown in Table 6, our method consistently mitigates hallucinations on LongFact, improving factual precision and reducing unsupported fact claims compared to the baseline. Although the improvement margin is smaller than in the FactScore experiments, this is expected since the baseline already achieves very high factual precision. Notably, the 1.4% precision gain from our method is comparable to the 0.9% improvement observed when moving from GPT-3.5-Turbo to GPT-4-Turbo, as reported in Wei et al.

Table 6. Performance of various variants of our method on the FactScore benchmark for Qwen3-32B model. No generation length constraint was applied. All variants were evaluated with the same hyperparameter settings described in Appendix B.1. Relative runtime is reported as a factor with respect to the runtime of the Following-Rewrite variant.

Sentence Sampling	Rewrite	Resp. (%)	Unsup.	Supp.	FPrec. (%)	Relative Runtime
Following	Yes	91.8	4.8	10.7	69.4	1.00
Temperature	Yes	95.6	4.9	9.0	64.8	1.01
Following	No	54.4	4.3	12.0	73.5	1.60
Temperature	No	40.1	2.3	7.4	76.2	1.55

(2024). We emphasize that the reported results are based on a single run without hyperparameter tuning due to the high API cost of evaluation, suggesting that further gains are likely achievable with hyperparameter optimization.

5.7. Ablation Study

This section presents an ablation study on two components of our method: sentence sampling and rewriting. As shown in Table 6, both Rewriting and the Following sampling strategy are critical for achieving strong performance. Enabling rewriting substantially boosts the response rate, while the Following strategy increases the number of supported fact claims and, when combined with rewriting, further improves factual precision.

We additionally report sensitivity to the hyperparameter, the number of sampled answers A in Figure 4, showing that performance is relatively stable across the evaluated range.

6. Limitations

We acknowledge two limitations in our method. First, prioritizing factuality may occasionally sacrifice informativeness, though our method empirically maintains a superior balance of these attributes compared to state-of-the-art baselines. Second, our method does not retrieve additional information from external resources, and thus cannot enable the model to answer questions for which it has no faithful knowledge. For example, if a model knows nothing about a person A, rejection and resampling alone will not produce new or accurate information. Nevertheless, significant value remains in the method’s ability to drastically reduce hallucinations in scenarios where the model possesses partial knowledge, alleviating the users’ effort in auditing the model output.

7. Conclusion

In conclusion, this work addresses the critical challenge of hallucinations in open-ended generation by introducing SHARS, a general inference-time compute framework that incrementally rejects hallucinated content and builds sub-

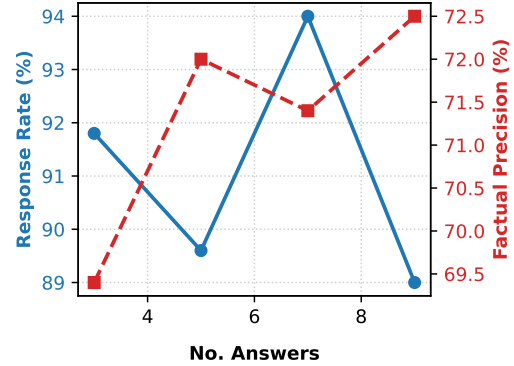


Figure 4. The change of response rate and factual precision with the number of answers, A .

sequent outputs upon verified information. Together with HalluSE, our improved uncertainty-based detection method, SHARS provides an effective and flexible approach to mitigating hallucinations while maintaining or enhancing informativeness. Extensive evaluations across multiple long-form factuality benchmarks demonstrate that our methods significantly advance the state of hallucination detection and mitigation, and importantly, reveal a promising inference-time scaling property of factuality. These findings highlight the potential of inference-time compute as a powerful and practical paradigm for improving the reliability of large language models, especially in high-stakes domains where accuracy and trustworthiness are paramount.

Acknowledgements

This work was funded under the Horizon Europe grant 101213369 DVPS. This study was also supported by the UK Medical Research Council, grant number MR/X029689/1.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Prompts

Prompts are given in Figure 5 and Figure 6.

B. Experiments

B.1. Configuration

FactScore (Min et al., 2023) evaluates LLMs by generating biographies for 182 individuals¹ (the labeled split), spanning diverse demographics and varying levels of rarity. Each generation is decomposed into atomic facts, which are verified against a reliable knowledge source, in this case a pre-saved Wikipedia. GPT-5 was used as the backend LLM for the benchmark. The prompts for query are given in Appendix A. The results are reported from a single run due to the high API cost of benchmark evaluation.

LongFact (Wei et al., 2024) is a benchmark for long-form factuality with two key differences from FactScore. First, it includes thousands of questions across 38 topics, though we used only a subset of 140 prompts following Zhao et al. (2025) due to resource constraints (evaluating a single generation costs at least \$0.19 (Wei et al., 2024)). Second, it relies on results from online Google Search rather than a pre-saved Wikipedia as the knowledge source. GPT-3.5-turbo-0125 is used as the backend LLM. The experimental setup follows that of FactScore in Section 5.1.

We describe here the hyperparameters of our method. The maximum number of tolerated consecutive hallucinated sentences sampling, N , is 10 across all setups. In Table 1, the number of probe questions, Q , the number of answers, A , and the semantic entropy threshold, θ , are 1, 3, 0.7 for Ours-Resp with Llama3.1-8B-Instruct; 3, 3, 0.3 for Ours-Info with Llama3.1-8B-Instruct; 2, 3, 0.3 for Ours-Prec with Llama3.1-8B-Instruct; 1, 5, 0.7 for Ours-Resp with Qwen3-32B; 2, 3, 0.2 for Ours-Info with Qwen3-32B; 2, 7, 0.6 for Ours-Prec with Qwen3-32B. In Table 2, the number of probe questions, Q , the number of answers, A , and the semantic entropy threshold, θ , are 1, 3, 0.5 for Ours-Info; 2, 3, 0.3 for Ours-Prec. In Table 5, the number of probe questions, Q , the number of answers, A , and the semantic entropy threshold, θ , are 3, 3, 0.3 for without length constraint; 2, 3, 0.5 for 200-words constraint. In Table 6, the number of probe questions, Q , the number of answers, A , and the semantic entropy threshold, θ , are 1, 3, 0.5 for all. All above hyperparameters, except for the ones for LongFact, are found through a coarse grid search.

Meanwhile, our method can deliver substantial performance improvements across diverse models and datasets using fixed hyperparameter values, and these gains can be further enhanced by tuning the hyperparameters for specific models or data. For example, we find that $Q=3$, $A=3$, and $\theta=0.3$ consistently achieve the highest or near-highest factual accuracy across all settings, while $Q=1$, $A=3$, and $\theta=0.5$ provide strong accuracy gains with moderate computational cost.

B.2. NLI Model in Semantic Entropy

The NLI model we used is DeBERTa V2 (He et al., 2021) with 900M parameters, identical to the one used in Semantic Entropy. Specifically, we used the deberta-v2-xlarge-mnli checkpoint from HuggingFace. It was trained on the following datasets: Wikipedia (English dump; 12GB), BookCorpus (Zhu et al., 2015; 6GB), OPENWEBTEXT (public Reddit content, Gokaslan & Cohen, 2019; 38GB), and STORIES (a CommonCrawl subset, Trinh & Le, 2018; 31GB). These sources span diverse domains, making the model effectively domain-agnostic.

C. Additional Results

C.1. Standard deviation

To assess robustness, we repeat the experiments for Qwen3-32B and Llama3.1-8B in Table 1 three times and measure the mean and standard deviation in Table 7, which confirm the robustness and consistency of our results.

We originally reported single-run results because the performance margins over baselines are large and stable across models, and the evaluation is expensive due to API costs. This is also consistent with common practice in prior work under similar constraints.

¹We exclude one individual named Focus... from the original dataset due to confusion with the band of the same name Focus and complications caused by special punctuation.

Table 7. Standard deviation of performance on the FactScore benchmark without constraints on response length.

Method	Resp. (%)	Unsup.	Supp.	FPrec. (%)
Llama3.1-8B				
Ours-Resp	99.27 $\pm$ 0.32	3.31 $\pm$ 0.10	5.60 $\pm$ 0.30	62.82 $\pm$ 1.55
Ours-Info	88.46 $\pm$ 0.00	1.82 $\pm$ 0.13	5.46 $\pm$ 0.41	75.01 $\pm$ 1.52
Ours-Prec	77.02 $\pm$ 1.22	1.46 $\pm$ 0.07	4.91 $\pm$ 0.12	77.02 $\pm$ 1.22
Qwen3-32B				
Ours-Resp	97.25 $\pm$ 0.95	5.55 $\pm$ 0.15	10.74 $\pm$ 0.62	65.90 $\pm$ 1.28
Ours-Info	91.58 $\pm$ 1.14	4.31 $\pm$ 0.22	11.33 $\pm$ 0.33	72.43 $\pm$ 1.16
Ours-Prec	82.97 $\pm$ 1.98	3.20 $\pm$ 0.26	10.89 $\pm$ 0.24	77.32 $\pm$ 1.36

D. Discussion

D.1. How does SHARS compare with retrieval-based methods?

SHARS and retrieval-based approaches (e.g., RAG) address complementary aspects of hallucination: RAG improves factuality by expanding the model’s knowledge, while SHARS improves how the model utilizes its parametric knowledge. Moreover, long-form factuality benchmarks often rely on external knowledge sources as ground-truth (e.g., FactScore uses Wikipedia), making direct comparisons between these approaches less meaningful.

Importantly, the two can be combined in multiple ways. First, SHARS can be used in an adaptive pipeline, where retrieval is triggered only when repeated rejection or abstention indicates insufficient knowledge, improving efficiency and reducing unnecessary retrieval noise. Second, SHARS can be applied on top of retrieval-augmented generation to filter or correct hallucinations caused by imperfect retrieval or intrinsic model errors.

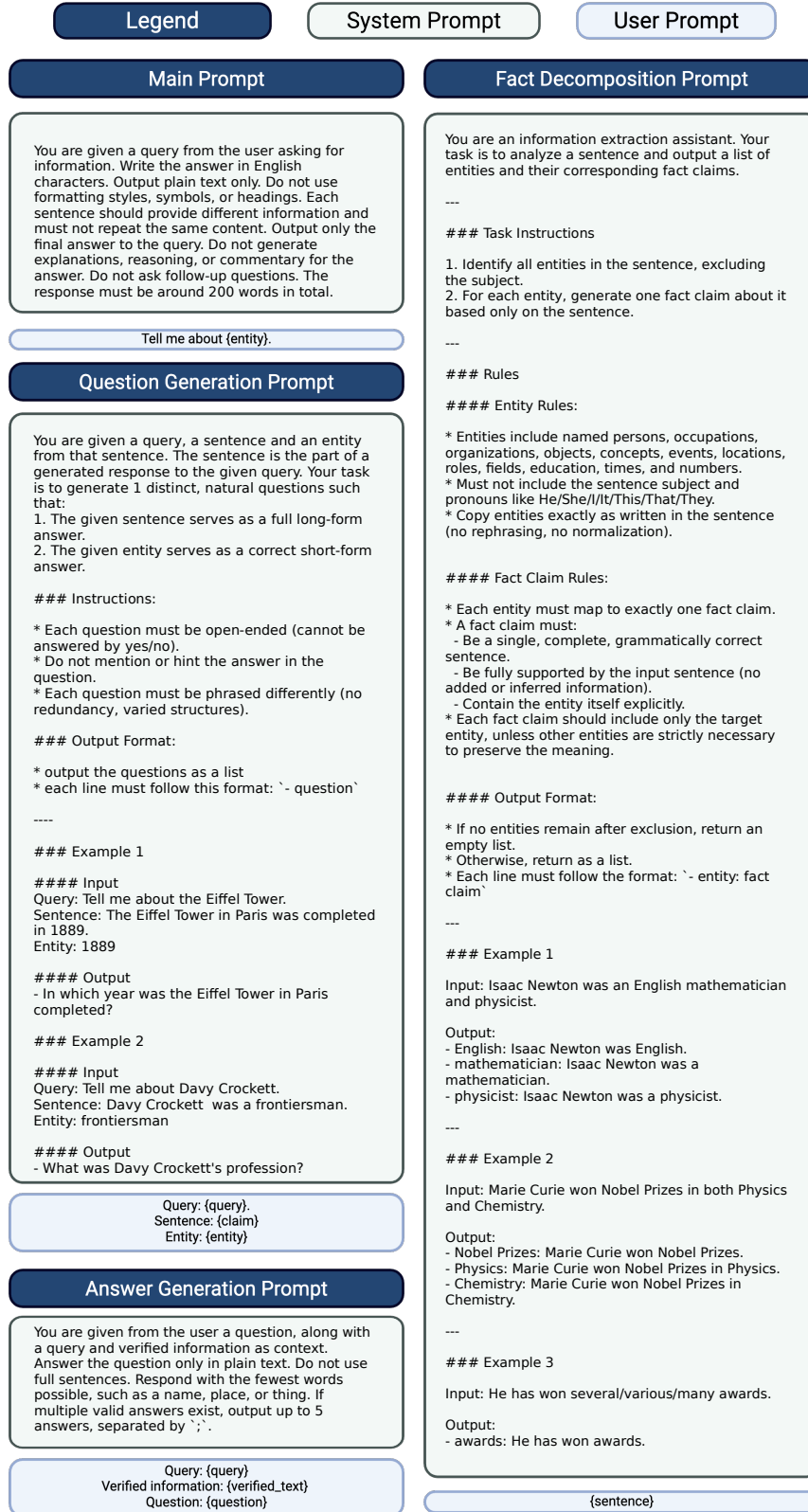


Figure 5. Set of prompts for various parts of the pipeline.

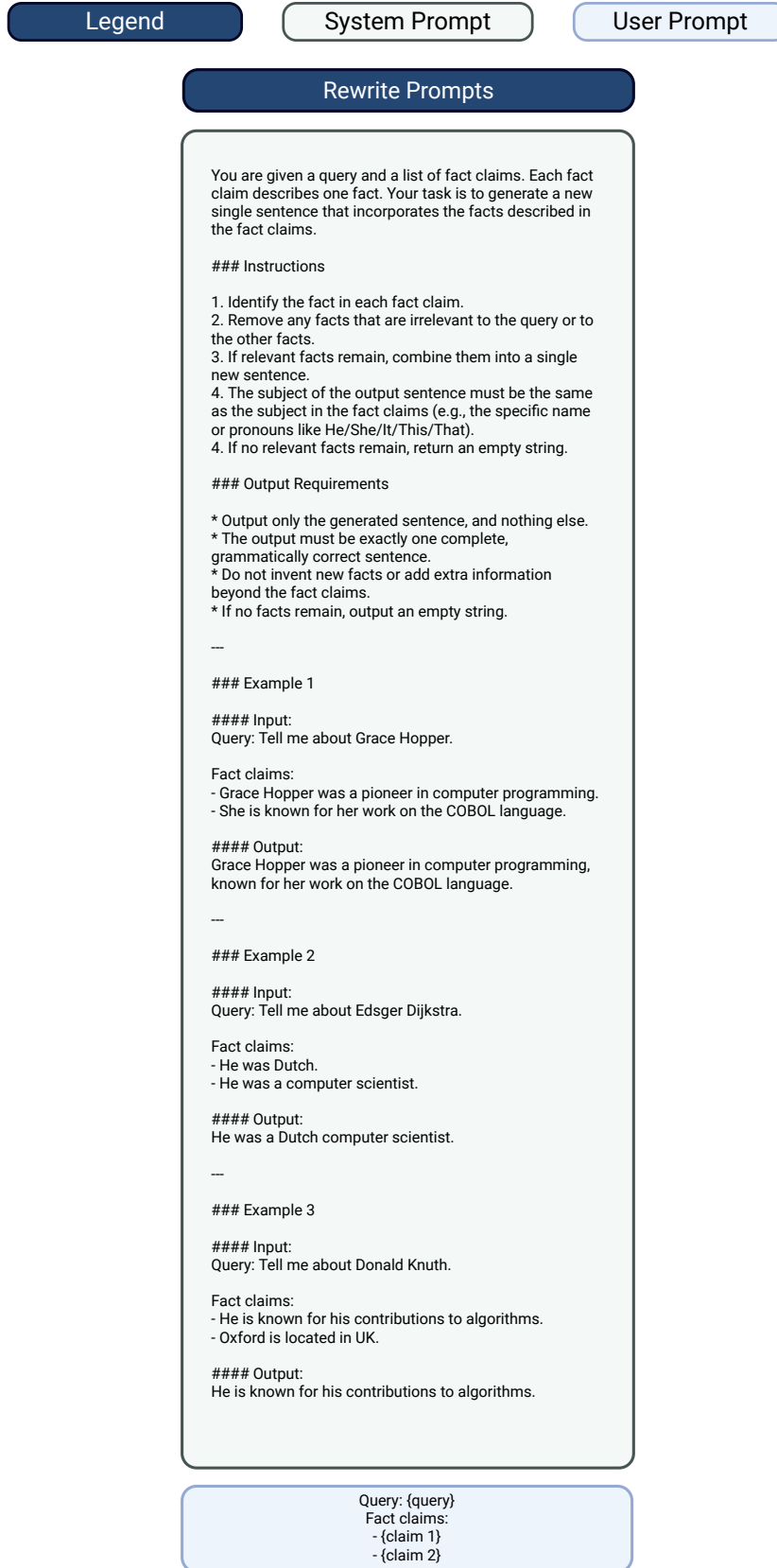


Figure 6. Additional prompt if rewrite is enabled.